Q1) Given the following data for this reaction: A+B+C ⟶ D

Experiment	[A]	[B]	[C]	RATE (M/s)
1	0.10 M	0.20 M	0.30 M	0.008
2	0.30 M	0.20M	0.30 M	0.216
3	0.10 M	0.20 M	0.50 M	0.0133
4	0.10 M	0.30 M	0.50 M	0.020

The rate law expression for the reaction is:

A. Rate = $k[A]^1[B]^1[C]^3$ B. Rate = $k[A]^1[B]^3[C]^1$ C. Rate = $k[A]^2[B]^1[C]^2$
D. Rate = $k[A]^3[B]^1[C]^2$ E. Rate = $k[A]^3[B]^1[C]^1$

Q2) What is the activation energy (in kJ) of a reaction whose rate constant increases by a factor of 35 upon increasing the temperature from 300 K to 360 K?

A. 27 B. 35 C. 42 D. 53 E. 69

Q3) The half-life for a first-order reaction (A→B) is 112.5 s. What was the original concentration if, after 2.0 minutes, the reactant concentration is 0.062 M?

A. 0.83 M B. 0.069 M C. 0.091 M D. 0.075 M E. 0.13 M

Q4) The half-life for the second order reaction: A→ B is 1570 seconds. What fraction of an initial amount of A remains after 23549 seconds if the initial concentration of A is 0.55M?

A. 0.333 B. 0.167 C. 0.051 D. 0.0625 E. 0.0313

Q5) How many times will the rate of the reaction ($3X + Y \rightarrow X_2Y$) change when the concentration of the substance Y is doubled and that of X is constant? **(Hint: the rate law is Rate (r) = k $[X]^3[Y]$)**

A. $r_2 = 24r_1$ B. $r_2 = 5r_1$ C. $r_2 = 2r_1$ D. $r_2 = 4r_1$ E. none

Q6) For the reaction: $2NO_{(g)} + O_{2(g)} \rightarrow 2NO_{2(g)}$ If ($\Delta S°_{rxn} = -177$ J $K^{-1}.mol^{-1}$, $\Delta G°_{rxn} = -120.7$ kJ/mol). Calculate the enthalpy of reaction in kJ per mole NO.

A. -43.3 B. -86.7 C. 43.3 D. 86.7 E. 173.4

Q7) Using the standard free energies of formation (kJ mol^{-1})

($NO_2(g)$, $\Delta G_f^o = +51.84$) ($NO(g)$, $\Delta G_f^o = +86.69$) ($SO_2(g)$, $\Delta G_f^o = -300.0$) ($SO_3(g)$, $\Delta G_f^o = -370.0$)

Calculate the standard free energy change, ΔG^o, for the reaction: $NO_2(g) + SO_2(g) \rightarrow NO(g) + SO_3(g)$

A. -35.15 kJ B.-104.9 kJ C-429.2 kJ D. -619.6 kJ E. -808.5 kJ

Q8) The reaction, A+ B → C is spontaneous at low temperatures but non-spontaneous at high temperatures. we can deduce that the values of ΔH ° and ΔS ° are constant over this temperature range then:

A. $\Delta H < 0$ and $\Delta S > 0$ B. $\Delta H > 0$ and $\Delta S < 0$ C. $\Delta H < 0$ and $\Delta S < 0$
D. $\Delta H > 0$ and $\Delta S > 0$ E. $\Delta G = 0$ at 298 K

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Q9) The Third Law of Thermodynamics states:

A. ΔE sys = q + w
B. The entropy of the universe increases for spontaneous processes.

C. $\Delta S = 0$ for a pure element in its most stable state at standard conditions.
D. $S = 0$ for a perfect crystal of a pure substance at 0 K.
E. none of the above

Q10) Which one of the following reactions has $\Delta S > 0$?

A. $C_{12}H_{20(l)} + 17\ O_{2(g)} \rightarrow 12\ CO_{2(g)} + 10\ H_2O_{(l)}$
B. $NH_4Cl_{(s)} \rightarrow NH_{3(g)} + HCl_{(g)}$
C. $2\ C_2H_{2(g)} + 5\ O_{2(g)} \rightarrow 4\ CO_{2(g)} + 2\ H_2O_{(s)}$
D. $Ba(OH)_{2(s)} + 2\ HCl_{(g)} \rightarrow BaCl_2{\cdot}2H_2O_{(s)}$
E. $(CH_3)_2CO_{(l)} + 4\ O_{2(g)} \rightarrow 3\ CO_{2(g)} + 3\ H_2O_{(l)}$

Q11) The value of $\Delta H°$ for the reaction below is -72 kJ. Calculate amount of heat released when 242.7 grams of HBr (M.w=80.9g/mol) is formed in this reaction. $H_{2\,(g)} + Br_{2\,(g)} \rightarrow 2HBr_{(g)}$

A. -144 B. -72 C. -108 D. -36 E. -72

Q12) A 1.5L of an ideal gas is expanded isothermally against external pressure equal to 2.96 atm at 25°C. During expansion, the gas absorbed 549 joule of heat. What is the volume of the gas after expansion?

A. 0.83 L B. 2.33 L C. 0.667 L D. 8.33 L E. 3.33 L

Q13) A correct description for the electron configuration of a vanadium atom ($_{24}Cr$) is

A. $[Ar]\ 4s^1\ 3d^5$ B. $[Ar]\ 4s^2\ 3d^3$ C. $[Ar]\ 4s^2\ 3d^5$ D. $[Ar]\ 3d^5$ E. $[Ar]\ 3s^2\ 3d^4$

Q14) Based on the information bellow, which expression represents the enthalpy for the following reaction:

$$C_3H_{8\,(g)} + 5\ O_{2(g)} \rightarrow 3\ CO_{2(g)} + 4\ H_2O_{(l)}$$

Reaction	Enthalpy of reaction
$C_3H_{8(g)} \rightarrow 3C_{(s)} + 4\ H_{2(g)}$	X
$CO_{2(g)} \rightarrow C_{(s)} + O_{2(g)}$	Y
$4H_{2(g)} + 2\ O_{2(g)} \rightarrow 4H_2O_{(l)}$	Z

A. $-X + 3Y + 2Z$ B. $X - 3Y + Z$ C. $-2X - 3Y + 2Z$ D. $-X - 3Y + 2Z$ E. $-X + 3Y - 2Z$

Q15) The combustion of titanium with oxygen produces titanium dioxide: $Ti_{(s)} + O_{2\,(g)} \rightarrow TiO_{2\,(s)}$

When 2.79 g of titanium (47.9g/mol) is combusted in a bomb calorimeter, the temperature of the calorimeter increases from 25.00 °C to 41.60 °C. If the heat capacity of the calorimeter equal 9.84 kJ/K. Calculate enthalpy of reaction (in KJ) for the combustion of a mole of Ti in this calorimeter.

A. -163.3 B. -165.8 C. -326.6 D. -3800 E. -3798

Q16) Which atom has the smallest first ionization energy?

A. $_{13}Al$ B. $_{19}K$ C. $_{31}Ga$ D. $_{12}Mg$ E. $_{11}Na$

Q17) Which one of the following statements is false?

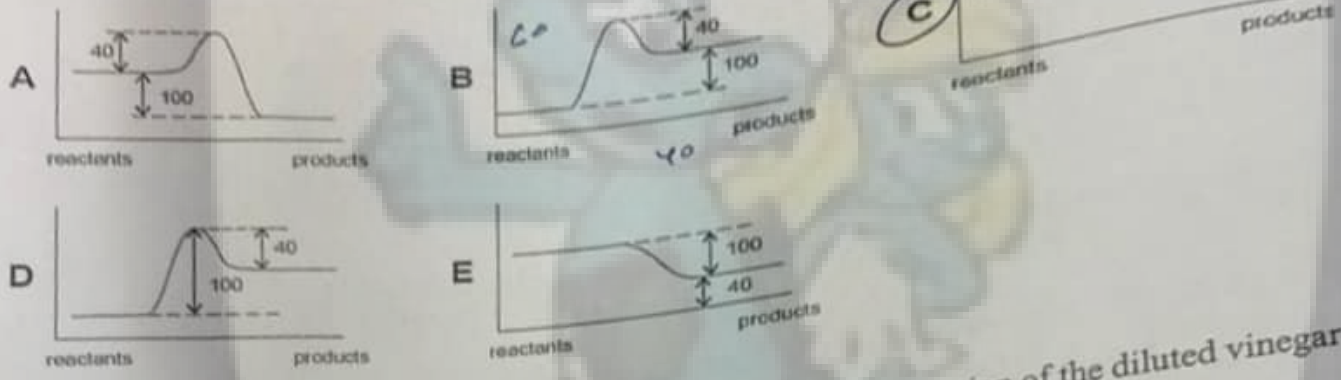
A. Internal energy is a state function..
B. Enthalpy is an intensive property.
C. Enthalpy is a state function.
D. The enthalpy change for a reaction depends on the state of the reactants and products.
E. The enthalpy of a reaction is equal to the heat of the reaction.

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Q18) What is the mass % of hydrogen in dimethylsulfoxide(C_2H_6SO)?(At.wt g/mol:C= 12, H=1,S =32,O =16)
A. 60.0 B. 20.5 C. 30.7 D. 7.69 E. 9.8

Q19) Under appropriate conditions, nitrogen and hydrogen undergo a combination reaction to yield ammonia: $N_{2\,(g)} + 3\,H_{2\,(g)} \rightarrow 2NH_{3\,(g)}$. The reaction percentage yield = 61.5% Calculate the actual yield of NH_3 produced if 7.1 g sample of N_2 are allowed to react with enough amount of H_2. (At.wts : N=14, H=1 g/mol)
A. 5.3 B. 6.7 C. 4.6 D. 1.5 E.17.4

Q20) An endothermic reaction (X↔Y) has enthalpy equal to 60kJ, and activation energy equal to 40 kJ for its reverse reaction. Which of one of the energy diagrams shown below best describes this reaction.



Q21) A 25.00 mL of a vinegar solution is diluted to 250.0 mL. If the concentration of the diluted vinegar solution is 0.05287 M. What is the concentration of the original vinegar?
A. 0.2578 M B. 0.5287 M C. 0.7825 M D. 0.8527 M E. 0.9978 M

Q22) If 50.0 mL of water is added to 50.0 ml of 0.60 M solution of $AlCl_3$. What is the final concentration of chloride ion (Cl^-) after adding water?
A. 1.8 M B. 0.3M C. 0.2M D. 0.6M E. 0.9 M

Q23) What is the molar concentration of a solution prepared by dissolving 10.0 g of cobalt(II) chloride dihydra ($CoCl_2 \cdot 2H_2O$ M.w=*165.87 g/mol*) in enough amount of ethanol to make a 114.6 mL solution .
A. 0.121 M B. 0.263M C. 0.526 M D. 0.734 M E. 0.972 M

Q24) For the reaction: $Na_2CO_{3(s)} + 2HNO_{3(l)} \longrightarrow 2NaNO_{3(s)} + CO_{2(s)} + H_2O_{(l)}$, how many grams of Na_2CO_3 (molar mass = 106.0 g/mol) are required to react completely with 152.2mL of 0.155 M HNO_3?
A. 0.125 g B. 0.205 g C. 0.410 g D. 1.250 g E.2.050 g

Q25) A 20 g of common salt is dissolved in 400 g of water. Calculate the mass percent concentration of th solution (w/w%).
A. 3.75 % B. 3.61 % C. 4.76 % D.7.45 % E. 5.88%

Good luck

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